

Shashank Modi

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EDUCATION

Vellore Institute of Technology <i>B.Tech Computer Science; CGPA: 9.4/10</i>	2022 – 2026 Vellore, India
D.A.V Sushil Kedia Vishwa Bharati <i>CBSE – 12th Grade; Percentage: 96.4%</i>	2022 Kathmandu, Nepal

WORK EXPERIENCE

Software Development Intern <i>Star Union Dai-ichi Life Insurance</i>	Jan 2026 – Present <i>Mumbai, Maharashtra</i>
- Optimized a high-volume stored procedure operating on large datasets, reducing query execution time by 2x and simplifying code by 3x through indexing, join restructuring, and use of temporary tables. - Developed a streaming XML export pipeline using .NET (XMLWriter, GZipStream, SqlDataReader), reducing memory usage to 5–20MB for large regulatory data exports. - Migrated legacy APIs from Python to .NET, aligning services with the organization's backend ecosystem. - Documenting legacy systems by creating system design documentation for undocumented applications.	
Software Engineer Summer Intern <i>Providence Global Center</i>	Jun 2025 – Aug 2025 <i>Hyderabad, Telangana</i>
- Developed a workflow management platform using React, ASP.NET Core, and MS SQL Server, replacing manual SharePoint-based processes across teams. - Designed and implemented REST APIs and automated reporting workflows using MailKit, improving process visibility and reducing manual effort.	
Technical Intern <i>Biomed International</i>	Jun 2024 – Jul 2024 <i>Kathmandu, Nepal</i>
- Redesigned product pages using HTML, CSS, and JavaScript, improving responsiveness and overall navigation. - Developed a secure patient portal and resolved user-reported issues, reducing recurring frontend bugs by 15%.	

PROJECTS

blip. Full-Stack PWA(Progressive Web App) <i>React.js, Node.js, PostgreSQL, Vercel</i>	🔗 Live Demo
- Engineered a rule-based NLP engine for real-time expense logging, achieving <100ms processing by eliminating LLM call latency. - Architected a social debt settlement system using PostgreSQL ACID transactions and row-level locking for safe concurrent balance updates. - Reduced perceived latency via optimistic UI updates and serverless warm-up strategies to mitigate cold starts. - Designed a high-contrast, minimalist interface optimized for rapid data entry and power-user workflows.	
Mental Fatigue Estimation <i>Python, scikit-learn, XGBoost, NumPy, Pandas</i>	🔗 View Code
- Built an ML pipeline to estimate cognitive fatigue levels (1–5) using keystroke dynamics and mouse telemetry data. - Engineered features such as typing latency and cursor movement patterns to convert raw input events into predictive signals. - Evaluated Linear Regression, Random Forest, and XGBoost models; achieved best performance with XGBoost (MAE: 0.93, MSE: 0.95) outperforming linear models by capturing non-linear behavioral patterns.	

SKILLS

Programming Languages: C++, Java, Python

Frameworks & Libraries: ASP.NET Core, React.js, scikit-learn, XGBoost, Material UI

Tools & Platforms: Git, Vercel, Render

Databases: MySQL, MS SQL Server, PostgreSQL (indexing, stored procedures, TVPs)

Core Concepts: Data Structures & Algorithms, Operating Systems, DBMS, OOP, Computer Networks

ACHIEVEMENTS

- Solved 250+ coding problems on LeetCode | [LeetCode Profile](#)
- Co-Secretary, Matrix Club (VIT): Conducted technical workshops and coordinated activities for 20+ members.